Audit of the SV efficiency proof, round 2: after the G1–G4 corrections

Source audited: `sv_efficiency_proof.md` (the source of `sv_efficiency_proof.tex`), rendered as `sv_efficiency_proof_md.pdf`, at the state of commit "Fix audit gap G4". Round 1 is in sv_proof_audit_2026-10.md. Not refereed.

Part A. Proof-check

A1. Claims and labelled assumptions

- **Theorem 2.** For $\ell\in\{\mathrm{Res}_A,-\tilde S\}$, the local and global efficiency hold in the dual norms over $H^1_*(\omega)=\{v=0\text{ on }\partial\omega\setminus\Gamma_N\}$, with $C=C(d,\vartheta)$.
- **Corollary 3:** $E\le\eta_E\le C(1+c_b)^2(E+\mathrm{osc})$.
- **Corollary 3':** $E_c\le\eta_{E,c}\le C(1+c_b)(E_c+\mathrm{osc})$, with $E_c=E+(M-1)|||e|||$ and $\eta_{E,c}=(M-1)\eta+\eta_E$.
- **Theorem 4:** the localised bounds; part 3 now carries $(1+c_b)^2$.
- **Proposition 5:** reliability, and efficiency at every iterate; part 3 is the content of the stopping rule.
- **Theorem 6:** variable ε, with the patch data $\Pi_{RT}(\psi_aD\nabla u_h)$ and $\hat r_\ell=r_\ell+\mathrm{div}_h(D\nabla u_h)$.

Assumptions: H1–H6 and A1–A6 as in round 1.

- **A2 (Neumann version)** is cited from [EV15], from memory.
- **A5** is Licht 2019; it is confirmed at abstract level only (see B3).

A2. Step table (only the steps that changed, and those that depend on them)

	constant	status				
	—	OK. Zero extension gives $H^1_*(\omega)\hookrightarrow H^1_D(\Omega)$, since $\partial\omega\cap\Gamma_D\subset\partial\omega\setminus\Gamma_N$				
	$N(d,\vartheta)$	OK				
		\psi_a			$	$\psi_a\in H^1_*(\om compatible
\in	$C(d,\vartheta)$	OK (G1 closed)				
the extension	1	OK				
	$\kappa\max\{1,c_b\}\le(2+c_b')c_b'$, with $c_b'=\max\{1,c_b\}$	OK (G2 closed)				
		e			+E$; Thm 1	—
		e			_{\oplus,c}$; Thm 2	—
	$1,\ 2M+1$	OK				
	$(1+c_b)^2$	OK				

	constant	status				
	exact	OK				
$\tilde S$	$C(1+c_b)^2$; $\eta_{\rm alg}\le\kappa C_\Pi C_I\|\mathrm{Res}_A(u^i)\|_*$	OK (G4 closed). Numerically: $\eta_E/E\in[2.9,6.0]$ from iterate 1				
	$1+\gamma$	OK				
	—	**N1**: not justified				
	exact	OK. **N3**: the compatibility claim needs consistent quadrature				
a	—	OK. **N2**: wording				
P_1^d$	$C(d,\vartheta)$	OK (G3 closed)				
	$C(d,\vartheta)\,\kappa_\varepsilon$	OK. Checked: $(\tilde g_a,w)-(\Pi_{RT}(\psi_aD\nabla u_h),\nabla w)=(\Pi_1(\psi_a\hat r_\ell),w)+\sum_F(\Pi_F(\psi_aj_F),w)_F$				
	$C(d,\vartheta)\,\kappa_\varepsilon$	OK				
	$C(d,\vartheta)\,\kappa_\varepsilon$	OK				
$ RTN$_1$	exact	OK. Code: `rt1_data`; data identical to $4\cdot10^{-16}$ when $D\nabla u_h$ is cellwise constant				

A3. Verdict

- **Correct, given the cited ingredients:** Theorem 2, Corollaries 3 and 3', Theorem 4, Proposition 5 parts 1–2, and Theorem 6. The cited ingredients are A2 with Neumann faces and A5. G1–G4 are closed.
- **Minor issues:** N1–N4 below. None affects a bound that is computed.

A4. Remaining issues and fixes

N1 (Proposition 5.3, last clause).

- **The problem.** "Further iterations can lower it by at most the share $\eta_{\rm alg}/(\eta_{\rm disc}+\eta_{\rm alg})\le\gamma/(1+\gamma)$" assumes that $\eta_{\rm disc}$ does not change. But $\eta_{\rm disc}(u^{i+k})$ also changes with the iterate, and nothing bounds it from below by $\eta_{\rm disc}(u^i)$.
- **Fix.** Replace the clause by: "the algebraic part is at most the fraction $\gamma/(1+\gamma)$ of the current bound."

N2 (Theorem 6.1, wording). "Reliability: Theorem 1 and Corollary 3 hold verbatim" should read "Theorem 1 and the upper bounds of Corollaries 3 and 3'". Corollary 3 itself is two-sided, and its lower bound is part 2.

N3 (Lemma V and quadrature).

- **The problem.** $\mathrm{Res}_A(\psi_a)=0$ at the discrete solution needs the assembly to integrate ε consistently: the assembled $\int_K\varepsilon$ must equal $\int_K\Pi_1\varepsilon=\int_K\varepsilon$. For ε not polynomial of low degree, the assembly quadrature and the estimator's degree-6 rule differ.
- **Effect.**
  - At the "discrete solution", δ_a carries a quadrature-consistency defect: in the smoke test, `sv_algebraic` and $\eta_{\rm def,A}$ differ by about 10^{-5} relative.
  - Reliability is unaffected, because the defect is counted in $\eta_{\rm def,A}$.
  - In the stopping split this defect is counted as algebraic.
- **Fix.** Either state "up to quadrature", or compute δ_a with the assembly quadrature.

N4 (H3).

- **The problem.** Theorem 2 does not use H3. Corollary 3 uses H3 as stated, $|c-\frac12\mathrm{div}\,b|_\infty\le c_b\sigma$, which excludes $\sigma=0$ with $c\neq\frac12\mathrm{div}\,b$. The code covers that case with the Friedrichs-based c_b (`friedrichs_constant`).
- **Fix.** Restate H3 as $c-\frac12\mathrm{div}\,b\ge\sigma\ge0$, with $c_b:=\min\{|c-\frac12\mathrm{div}\,b|_\infty/\sigma,\ |c-\frac12\mathrm{div}\,b|_\infty C_F^2/\varepsilon\}$. This was noted in round 1 under optimality and is still open.

A5. Optimality

- **Hypotheses actually used:**
  - H1, A1, A3 and A6 throughout;
  - H2 (or its §11 replacement) in Lemmas B, D and G;
  - H3–H4 only in Remark 1.2, Corollaries 3/3', Theorem 4.3 and Proposition 5.2;
  - H6 through the cellwise-constant (or, for §11, P1) D;
  - A2 in Lemma G case 2; A4 in Lemma H(ii) and Lemma L;
  - A5 in Lemma H(ii) (the second claim) and in Proposition 5.2 (the bound on $\eta_{\rm alg}$) only.
- **Sharpness in c_b.**
  - Corollary 3' has constants 1 and $2M+1$ between E_c and $|\mathrm{Res}|_*+|\tilde S|_*$; both are attained up to a constant by the triangle inequalities used.
  - The smoke effectivity is $\approx1.2\,c_b$ for $c_b\le10^3$.
  - Linear dependence is therefore the best possible in $|||\cdot|||_{\oplus,c}$ with a constant σ.
- **A sharper track for $c_b\gg1$** (the c_b-free one):
  - **Idea.** Put the variable reaction weight into the norm, $|||v|||_c^2=\varepsilon|\nabla v|^2+|(c-\frac12\mathrm{div}\,b)^{1/2}v|^2$. Then $|B_S'(w,v)|\le|||w|||_c|||v|||_c$ exactly, so $c_b=1$ and $\kappa=3$, with no reaction-dominance factor.
  - **Cost.** The patch problems need a cellwise σ_K (the SV20 construction with variable κ^2, as in Ainsworth–Vejchodský 2014 for reaction–diffusion [from memory, verify]). The defect bound needs a cellwise σ. Efficiency then carries the patch contrast of $c-\frac12\mathrm{div}\,b$, which is local, like κ_ε in Theorem 6.
  - **Payoff.** For AJ27 Ex. 4.3 this removes the factor $\approx8\cdot10^4$. This is the most useful extension of the analysis.
- **Proposition 5.** Parts 1–2 are the strongest statements this route gives. The Ern–Vohralík-type comparison $E(u^i)\lesssim E(u_h)$ needs a discrete stability of the AFC operator. That is open, and in general false without monotonicity.

- **Theorem 6.** Efficiency with the patch contrast κ_ε is optimal for local estimators when ε is not quasi-monotone: Li 2026 shows $C_{\rm rel}C_{\rm eff}\gtrsim M^{1/4}$. Under quasi-monotonicity, contrast-robustness would need harmonic patch weights. This is not attempted.

Status of the fixes (18 Oct 2026)

- **N1:** fixed. Proposition 5.3 now says that the algebraic part is at most γ/(1+γ) of the current bound, and nothing about later iterates.
- **N2:** fixed. Theorem 6.1 now refers to the upper bounds of Corollaries 3 and 3'.
- **N3:** fixed in the proof and the code.
  - Lemma V splits the defect δ_a = δ^alg + δ^quad.
  - `EstimatorStop(kind="sv")` now takes η_alg from the assembled residual (`sv_algebraic`) also at the full evaluations, so the quadrature part stays in η_disc. The bound η_disc + η_alg is unchanged.
  - Proposition 5.2 carries the extra term osc_q in the bound on η_alg.
- **N4:** fixed. H3 is restated as c − ½div b ≥ σ ≥ 0 with c_b = min{‖·‖_∞/σ, ‖·‖_∞C_F²/ε}, matching `friedrichs_constant`. Remark 1.2 shows both entries.

Part B. Literature scan (round 2)

B1. Answer

Nothing new since round 1 changes the novelty assessment or the track.

- **No competitor.** No 2026 work combines algebraic stabilisation with guaranteed equilibrated estimates; the newest AFC estimator work (Ahmed–Jha, arXiv 2602.14504) is residual-based.
- **Prior art for the G3 construction.** No source states $\Pi_{RT}(\psi_aD\nabla u_h)$ explicitly. It is the natural RTN analogue of the patch data in Cai–Cai–Zhang (Numer. Math. 2020) and Ainsworth–Vejchodský, so it should be presented as standard, not new.
- **New neighbour.** A 2026 ESAIM: M2AN paper on CutFEM with discontinuous coefficients (doi 10.1051/m2an/2026043) uses equilibrated fluxes in an immersed RT space and tracks the coefficient dependence of the efficiency constant. I have seen only the search snippet; it is a neighbour, not a competitor.
- **Track.** It is right. The one better track for strongly variable reaction is the variable-weight energy norm in A5.

B2. Frontier changes since round 1

result	scheme	estimate	hypotheses	mechanism	uniform in
ESAIM: M2AN (2026), CutFEM, discontinuous coefficients	unfitted FEM, interface	equilibrated, guaranteed; efficiency tracked in the coefficients	snippet only	immersed RT fluxes	h (coefficient dependence explicit) [verify]
galerkit Cor. 3'	any conforming P1 algebraic stabilisation	$E_c\le\eta_{E,c}\le C(1+c_b)(E_c+\mathrm{osc})$	H1–H6	weighted augmented norm	ε, σ, h; linear in c_b
galerkit Prop. 5.2	any iterate of any solver	two-sided in $E(u^i)$	A5 for $\eta_{\rm alg}$	Thm 2 at the iterate	ε, σ, h, iteration index

Unchanged from round 1: Li 2026 (impossibility without quasi-monotonicity), Licht 2019, Ahmed–Jha 2026, Jha 2021/2024, Verfürth 2005, Cai & Cai 2025, SV20, ESV10, ABR17, EM16, Ern–Vohralík 2013, HPSV21.

B3. The four checks

1. **A5 (Licht 2019).** The abstract (arXiv 1710.06868; Math. Comp. 88 (2019) 607–635) says the smoothed projections commute with the exterior derivative, preserve homogeneous boundary conditions on a fixed boundary part, and have uniform bounds for shape-regular meshes. The norm of those bounds (L^2, as A5 needs, or L^p) is not stated in what I could access; the preprint itself was unreachable from this environment. **Status: plausible, not verified.** Read Theorem 1 of the preprint before submission.

2. **Ern–Vohralík 2013 and the γ condition.** The accessible material (conference and lecture abstracts) confirms guaranteed upper and robust lower bounds with adaptive stopping. It does not show where the smallness of γ enters. The remark in Proposition 5 stays "[from memory, verify]".
3. **c_b.** I found no estimator that is robust in $|c-\frac12\mathrm{div}\,b|_\infty/\sigma$ in the augmented norm with a constant σ. The variable-weight norm is the known way around it (functional/Repin-type estimates, arXiv 1601.05310).
4. **AFC stopping and AFC equilibration (2025–2026).** Nothing new found.

B4. Gaps ranked by novelty-to-effort

1. **N1–N4:** wording and quadrature. Low effort; needed for correctness of the text.
2. **The variable-weight energy norm** (A5, sharper track). Medium effort: patch problems with a cellwise σ, a cellwise defect bound, and Lemma G with θ_K cellwise. It removes c_b and is the most useful extension.
3. **Read Licht 2019, Theorem 1**, to close A5. Low effort, but it needs network access to the preprint.
4. **The stopped iterate against the discrete solution.** High risk; open for non-monotone AFC.

B5. References (new or rechecked in round 2)

- M. W. Licht, Smoothed projections and mixed boundary conditions, Math. Comp. 88 (2019) 607–635, arXiv 1710.06868, preprint. [abstract]
- M. W. Licht, Smoothed projections over weakly Lipschitz domains, arXiv 1612.02756. [abstract]
- CutFEM with discontinuous coefficients, ESAIM: M2AN (2026), doi 10.1051/m2an/2026043. [search snippet only]
- Cai, Cai, Zhang, higher-order conforming equilibration, Numer. Math. 2020 (preprint). [abstract]
- Ainsworth–Vejchodský and related equilibration, see arXiv 1812.06678 for the SV20 context. [read]
- A. Ern, M. Vohralík, SISC 35 (2013), HAL; M. Vohralík, Cargèse lecture notes. [abstracts]
- N. Ahmed, A. Jha, arXiv 2602.14504 (2026); A. Jha, arXiv 2404.02804. [abstracts]
- Eigel–Merdon, J. Sci. Comput. 67 (2016); Tobiska–Verfürth, arXiv 1402.5892; Cai & Cai, arXiv 2107.06341. [read earlier]